

Assignment 2 - Discrete Mathematics (Monsoon 2025)

*“Solving problems is a practical art, like swimming, or skiing, or playing the piano: you can learn it only by imitation and practice”
- by George Pólya*

Submission deadline: 11:59 PM on 28th October, 2025.

Maximum marks: 60

Instructions. Consider the following while preparing the answers for the given questions.

1. Kindly go over the academic dishonesty policy of IIIT Delhi given on [this link](#). Any deviation from this policy will lead to the strictest possible punishment.
2. Each question is of 10 marks.
3. It is prohibited to use AI-related tools like chatGPT to solve the questions, unless explicitly mentioned. Any violation in this regard will be handled very strictly.
4. You are encouraged to write solutions in $\text{\LaTeX}$. Anyone who submits homework in $\text{\LaTeX}$ (i.e., in the pdf format and not the source code) will get an additional 20% of their score in that homework.
5. While writing the solutions, you can refer to the topics, theorems, examples, etc., taught in the class. It is not allowed to refer to anything related to propositional and predictive logic not covered in the class.
6. Late submissions are not allowed.
7. If you discuss anything related to the solution of a question in this assignment with anyone, you must mention their names.

1 Problem Set

Question 1.1. 1. (6 marks). Suppose there are finitely many prime numbers $p_1, \dots, p_n$ for some natural number $n \geq 2$. Prove or disprove the following: for every $I \subseteq \{1, \dots, n\}, J \subseteq \{1, \dots, n\}$ satisfying $I \neq \emptyset, J \neq \emptyset, I \cap J = \emptyset$ and $I \cup J = \{1, \dots, n\}$, the natural number $\prod_{i \in I} p_i + \prod_{j \in J} p_j$ is prime. Here, $\prod_{i \in I} p_i, \prod_{j \in J} p_j$ mean the product of prime numbers whose indices are in I and J , respectively.

2. (4 marks). Let A be a finite set and R be an equivalence relation on A . Prove that

$$|A| = \sum_{[a]_R \in A/R} |[a]_R|,$$

where $\sum_{[a]_R \in A/R} |[a]_R|$ means the sum of cardinalities of all equivalence classes of A with respect to R .

Question 1.2. 1. (5 marks). Read about Russell's paradox online, thoroughly understand it, and then write it in your own words.

2. (5 marks). Let $n \in \mathbb{N}, n > 1$ be arbitrary. Prove or disprove the following: there exists a propositional formula u in propositional variables $p_1, \dots, p_n$ such that

$$\left| \{(a_1, \dots, a_n) \in \{T, F\}^n : u(a_1, \dots, a_n) = T\} \right| = \frac{2^n}{2} + 1,$$

where T, F means true and false, and $\{T, F\}^n$ is the set obtained by taking the Cartesian product of $\{T, F\}$ by itself n times. Here $u(a_1, \dots, a_n)$ means that we substitute $p_i = a_i$ in u for every $i \in \{1, \dots, n\}$.

Question 1.3. Prove or disprove each of the following:

1. (2.5 marks). Let A, B be sets. Then, $\text{pow}(A \cup B) = \text{pow}(A) \cup \text{pow}(B)$.
2. (5 marks). Let A be a non-empty set and let R, S be two equivalence relations on A . Then, $R \cap S$ and $R \cup S$ are also equivalence relations on A .
3. (2.5 marks). Let A, B be non-empty sets and $F : A \rightarrow B$ be a function. Let R be a relation on A defined as follows: for every $a, b \in A$, $(a, b) \in R$ if and only if $F(a) = F(b)$. Then, R is an equivalence relation.

Question 1.4 (10 marks). Let A be a non-empty set and R be an arbitrary equivalence relation on A . Let A/R be the set of all equivalence classes of A with respect to R , and $G : A \rightarrow A/R$ be the function such that for every $a \in A$, $G(a) := [a]_R$. Let B be a non-empty set and $F : A \rightarrow B$ be an arbitrary function that satisfies the property that if $(a, b) \in R$ then $F(a) = F(b)$.

Prove that there exists a unique function $H : A/R \rightarrow B$ such that $F = H \circ G$. Here, $=$ refers to the equality of functions.

Question 1.5. 1. (3 marks). Let A, B be non-empty sets and $F : A \rightarrow B$ be a function. For every $y \in B$, $F^{-1}(y) := \{x \in A : F(x) = y\}$. Let $P := \{F^{-1}(y) : y \in B\}$. Does P form a partition of A ? Prove your answer.

2. (7 marks). Is $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$ a countably infinite? Prove your answer.

Question 1.6. 1. (4 marks). Read about partial-order relations and partial-ordered sets online and describe them in detail in your own words.

2. Let A, B be non-empty sets. Then, prove or disprove the following.

- (a) (3 marks). Let $F : A \rightarrow B$ be a function. Then, F is injective if and only if for every $X \subseteq A, Y \subseteq A$, $F(X \cap Y) = F(X) \cap F(Y)$. Here, for every $Z \subseteq A$, $F(Z) := \{F(z) : z \in Z\}$.
- (b) (3 marks). Let $F : A \rightarrow A$ be a function that satisfies $F(F(x)) = x$ for every $x \in A$. Then, F is bijective.

Appendix

2 Some Results

Fact 2.1. Let a be a natural number which is not prime. Then, there exists a prime number b such that $b < a$ and $b \mid a$.

Fact 2.2. Let $p_1, \dots, p_n, q$ be prime numbers. If $q \mid p_1^{e_1} \cdots p_n^{e_n}$, where every e_i is a non-zero natural number, then there exists a prime number p_i for $1 \leq i \leq n$ such that $q \mid p_i$. Here, $p_i^{e_i}$ means the number obtained by multiplying p_i by itself e_i many times.

Fact 2.3. Let p be a prime number and a be a natural number. Let $k > 0$ be a natural number. If $p \mid a^k$ then $p \mid a$.